

OVIEDO, Spain

WMO: 080150

Lat: 43.3533N

Lon: 5.8742W

Elev: 1112

StdP: 14.11

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.4	(c) 34.4	(d) 26.8	(e) 21.6	(f) 37.2	(g) 29.3	(h) 24.3	(i) 39.2	(j) 23.4	(k) 53.2	(l) 20.6	(m) 51.9	(n) 4.2	(o) 170	(p) 0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.5	(c) 81.2	(d) 68.8	(e) 77.8	(f) 67.2	(g) 75.2	(h) 65.8	(i) 70.2	(j) 78.6	(k) 68.4	(l) 75.6	(m) 67.0	(n) 73.4	(o) 5.8	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	104.6	74.2	65.7	98.9	71.9	64.6	95.1	70.3	34.8	78.3	33.3	75.8	32.1	73.9	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.6	(b) 13.9	(c) 11.3	(e) 29.0	(f) 88.9	(g) 2.2	(h) 4.2	(i) 27.4	(j) 91.9	(k) 26.1	(l) 94.3	(m) 24.9	(n) 96.7	(o) 23.3	(p) 99.7		
(a) 17.6	(b) 13.9	(c) 11.3	(e) 27.8	(f) 74.7	(g) 2.3	(h) 2.7	(i) 26.1	(j) 76.7	(k) 24.8	(l) 78.2	(m) 23.5	(n) 79.8	(o) 21.8	(p) 81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	47.2	47.6	51.0	52.7	57.2	62.4	65.6	66.4	63.5	59.2	51.1	48.1
	DBStd	8.72	5.90	5.96	6.09	5.45	5.35	4.95	3.98	3.99	4.43	5.36	5.46	5.93
	HDD50	490	124	105	61	32	7	0	0	0	0	4	49	108
	HDD65	3480	550	487	436	369	249	107	40	30	81	191	416	524
	CDD50	2701	39	38	91	114	230	372	485	507	404	289	83	49
	CDD65	215	0	0	1	1	6	29	60	72	35	11	0	0
	CDH74	915	0	0	2	14	45	176	223	280	133	41	1	0
	CDH80	174	0	0	0	2	6	47	32	62	20	5	0	0
(14) Wind	WSAvg	4.5	5.1	5.2	5.3	4.8	4.2	4.1	3.9	3.8	4.0	4.0	4.7	4.7
(15) Precipitation	PrecAvg	40.7	4.0	3.7	3.5	4.0	3.2	2.6	1.9	2.2	2.3	3.8	5.3	3.5
	PrecMax	53.3	8.3	9.2	7.6	10.0	8.8	8.7	5.3	7.3	4.2	8.3	8.9	7.0
	PrecMin	31.2	0.6	0.6	0.2	0.1	1.0	0.6	0.4	0.6	0.8	0.7	0.9	0.2
	PrecStd	6.3	2.1	2.3	2.1	2.4	1.7	1.9	1.2	1.6	1.0	2.0	2.0	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	67.5	72.4	75.3	79.1	86.6	83.8	86.2	82.7	78.7	69.2	65.5
		MCWB	56.1	54.5	56.8	61.0	65.5	70.4	71.1	71.7	67.9	64.0	58.7	53.9
	2%	DB	62.1	62.7	67.5	69.0	73.4	78.5	79.2	80.3	77.4	73.7	65.3	62.3
		MCWB	53.4	52.8	55.8	58.1	63.1	67.0	68.8	69.1	67.1	62.8	57.4	53.0
	5%	DB	59.2	59.5	64.1	65.4	69.5	74.2	76.1	76.4	74.1	70.3	62.1	59.7
		MCWB	51.5	51.1	53.9	56.4	61.0	65.0	67.0	67.2	65.3	61.4	55.7	51.9
	10%	DB	56.3	56.6	60.6	61.8	66.0	70.9	73.2	73.7	71.2	67.4	59.2	57.2
		MCWB	49.9	49.7	52.2	54.5	58.9	63.3	65.6	65.9	64.0	60.1	53.8	50.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	57.2	58.9	62.9	67.0	71.2	72.5	73.5	70.2	66.9	61.2	56.2
		MCDB	64.1	63.6	68.3	73.3	75.8	83.4	81.5	84.0	78.6	73.7	66.6	60.9
	2%	WB	54.7	54.0	56.7	59.2	63.8	68.1	69.4	70.4	68.0	64.4	58.5	54.4
		MCDB	60.6	60.4	65.4	67.3	72.4	77.3	77.1	78.1	75.5	70.7	63.1	59.6
	5%	WB	52.4	52.1	54.7	57.2	61.6	66.0	67.8	68.3	66.3	62.7	56.6	53.0
		MCDB	57.6	57.7	62.2	63.8	68.3	73.0	74.6	74.4	72.6	68.1	60.7	58.3
	10%	WB	50.5	50.5	53.0	55.2	59.7	64.0	66.3	66.8	64.7	61.1	54.7	51.4
		MCDB	55.0	55.4	59.8	60.9	65.1	69.6	71.9	72.2	69.8	66.0	58.4	56.4
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	12.1	13.7	13.9	13.5	13.1	12.9	13.5	13.9	13.1	10.7	10.9
		MCDBR	14.2	16.8	19.1	19.6	19.2	19.6	18.0	18.8	18.5	17.4	15.0	13.5
		MCWBR	9.4	10.4	10.6	10.8	10.4	10.3	8.9	9.3	9.6	9.4	9.7	8.6
		MCDBR	12.8	14.8	17.3	17.9	17.8	17.7	16.6	16.9	17.0	15.3	13.4	12.0
	5% WB	MCWBR	9.1	9.9	10.4	10.4	10.2	9.9	8.7	8.9	9.2	9.0	9.2	8.3
(40) Clear-Sky Solar Irradiance	taub		0.314	0.324	0.359	0.377	0.391	0.399	0.388	0.385	0.368	0.350	0.325	0.309
	taud		2.514	2.473	2.383	2.342	2.329	2.328	2.361	2.380	2.424	2.483	2.513	2.522
	Ebn at Noon		261	278	279	282	280	277	279	276	272	264	254	252
	Edh at Noon		22	27	34	38	39	40	38	36	32	27	22	20
(44) All-Sky Solar Radiation	RadAvg		466	725	1094	1451	1715	1853	1890	1711	1367	879	525	425
	RadStd		52	101	142	145	139	140	109	110	54	82	63	52

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page